

Beyond Dexterity:

On the Engineering of Dexterous Hands for Real-World Applications

Why do dexterous hands fail in the real world? Unlike parallel grippers, they pack dozens of precision components — joint encoders, tactile sensors, and high-reduction actuators into a compact, anthropomorphic package. This density creates multiple failure points. In practice, they are unavoidably subjected to impacts, off-axis over-torquing, and shock loads that strip gears, fracture sensor traces, or stall motors into overheating—turning a promising research platform into an expensive paperweight after just hours of unstructured use.

In this talk, I will present how we tackled these challenges head-on with the Apex Hand, engineered not just for dexterity, but for durability under real-world punishment. Join me to explore how we are bridging the gap between human-like manipulation and reliable, day-in-day-out operation.



Dr. Wang Wei Lee

Co-Founder, CTO

Wang Wei Lee is Co-founder of Rysen Robotics. He holds a PhD from the National University of Singapore and previously led tactile sensing R&D at Tencent Robotics X. His work on tactile sensors has been published in Science and Nature series journals, with close to 2,000 citations. He founded Rysen Robotics in December 2024 to build dexterous hands that work outside the lab.

Simulation models and SDK:

<https://github.com/RysenRobotics/>

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